

We built a scanning tunnelling microscope in a basement

Jacob Katz — University of California San Diego
Nuh Shaheer — Worcester Polytechnic Institute

Self-directed. No faculty advisor, no grant, no laboratory. Eleven weeks of planning and design, eight weeks of building. First power-on 29 August 2026; first bench session 19 September 2026. 27 session logs, 163 raw data files, 92 photographs of our own hardware and every analysis script are public — including the results we later withdrew.

We detected tunnelling, but weren't able to maintain tunnelling range for long enough to get an image.

Every subsystem a tunnelling microscope needs is built, working and calibrated. The amplifier resolves picoamps, the chain agrees with theory to **0.13 of a standard error**, and a real junction responds to voltage and to height.

0.13 σ

The chain agrees with theory

Measured gain against what Ohm's law required, worked out before the run. 53 readings. MEASURED 2026-09-16

4 pA

Amplifier's own input current

Against the ~ 1 nA it has to find ~ 250 times smaller. MEASURED 2026-09-15

16–59 M Ω

A real junction, made and swept

Resistance falls as the voltage rises — not a short, not an open. MEASURED 2026-09-17

9–19

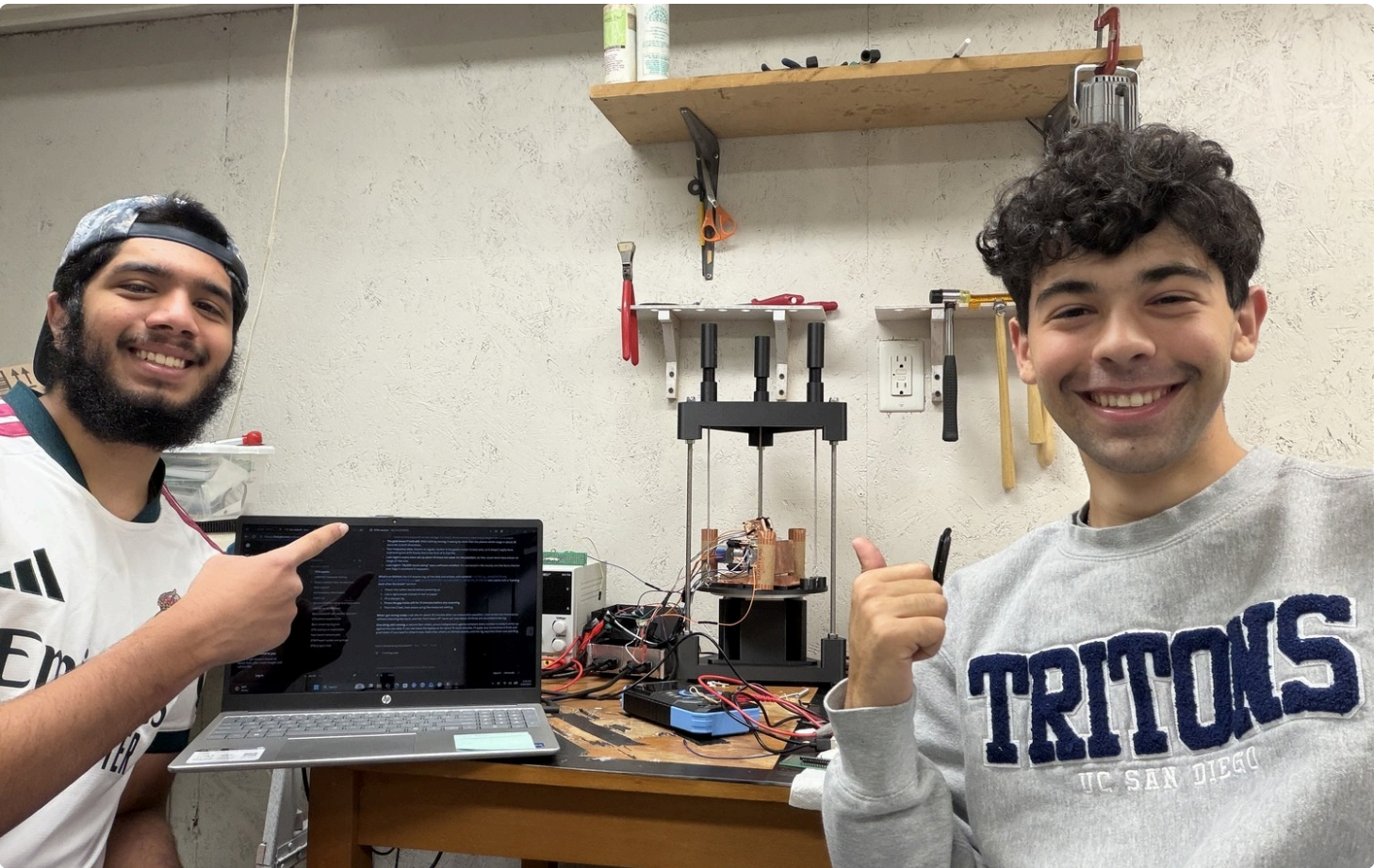
Counts of noise with a junction live

Our electronics are not what limits us, and we proved it with numbers. MEASURED 2026-09-19

60,928

Readings inside the tunnelling window

Taken across 109 separate approaches, with bias on the whole way. MEASURED 2026-09-19



Nuh Shaheer, left, Jacob Katz, right. The morning of the last bench session, with the instrument still assembled behind them. Identification and permission to publish SAID by Jacob, 2026-09-19 (Nuh, second-hand via Jacob, 2026-09-20). IMC_B633.

Two undergraduates. A basement. Our first soldering.

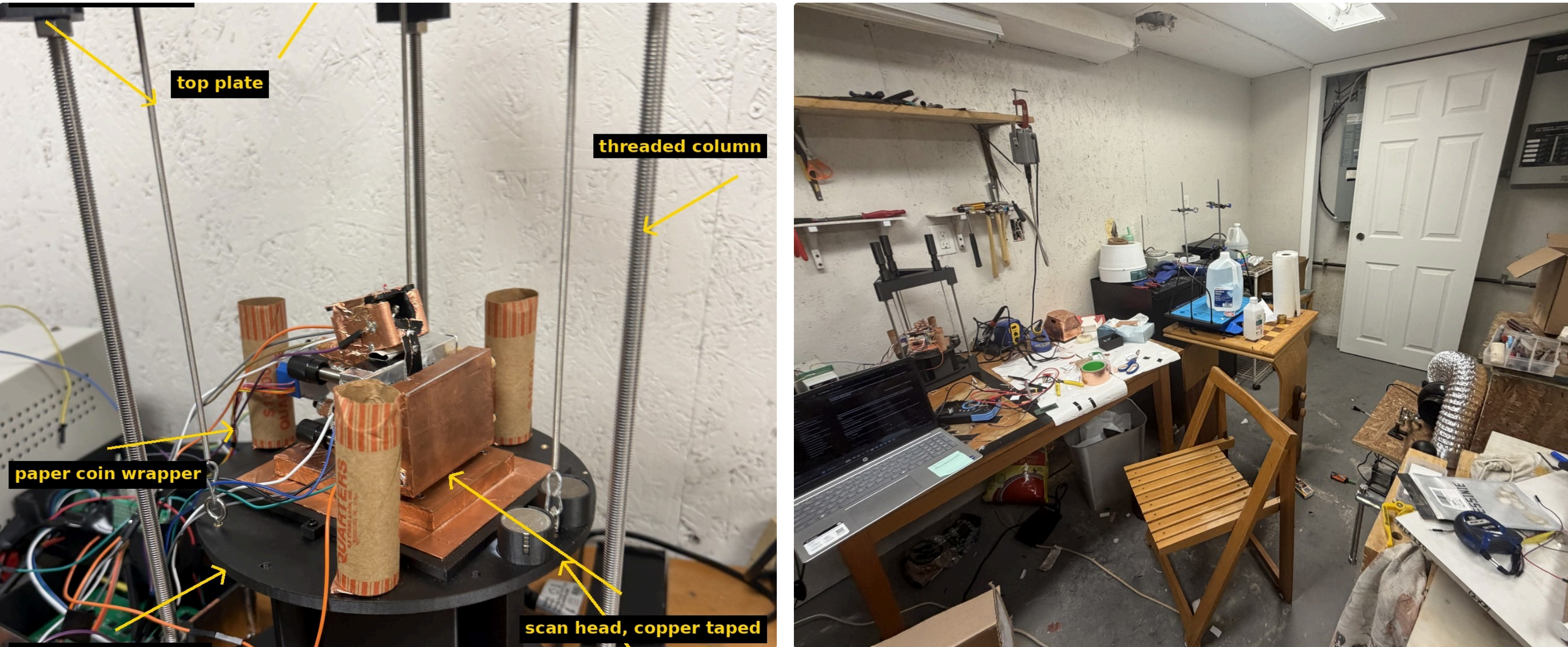
Neither of us writes code, and neither had much electronics experience — **this project was the first real soldering either of us has done**. It was built and run in the basement of a house people live in, with people moving around and no control over the air or the temperature. A wooden bench on a concrete floor, a few meters from the breaker panel. **Not a lab. Not a clean room.**

This is not a disclaimer. It is the result. A tunnelling microscope is normally a clean-room instrument — and the blocker we found and measured is exactly the class of problem those rooms exist to remove. Stated up front, our limitation becomes a measurement of what this environment costs you.

Conditions stated by Jacob Katz, 2026-09-19.

01 We built the whole thing

There was no kit. We designed it, ordered the boards, printed the frame, wrote the firmware, and brought up every stage on its own before trusting the next one.

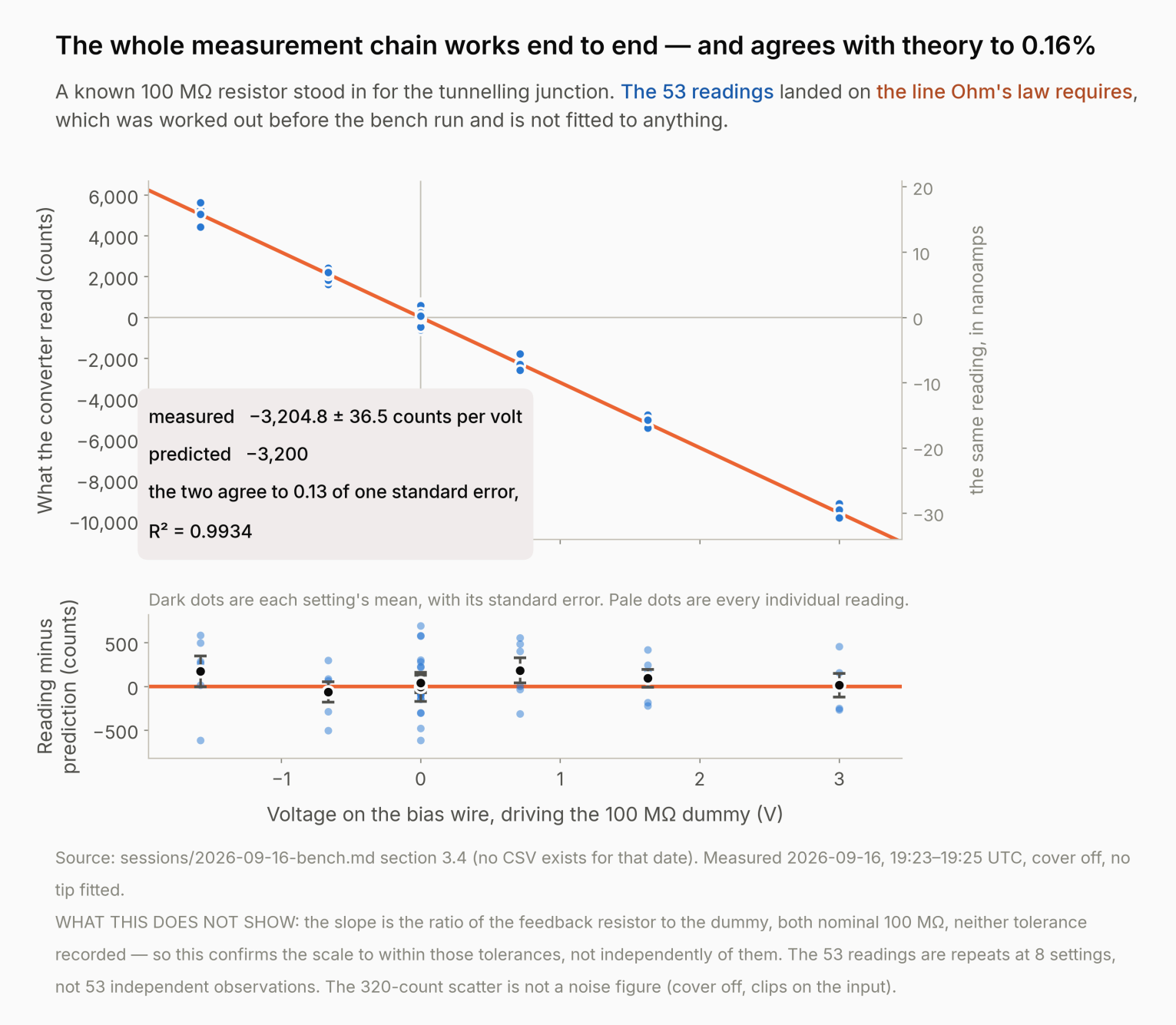


Left: the instrument on the morning it came apart to be moved. Printed frame; springs down to the suspended platform; copper-taped scan head on a copper ground plane. Right: where it happened — a wooden bench on a concrete basement floor, a few meters from an electrical panel. READ, 2026-09-19.

A Tensy 4.1 drives four 16-bit converters over a 1 MHz bus and a 26-way ribbon — X, Y and sample bias at ± 3 V, Z at ± 10 V — moving a piezoelectric disc that carries a cut tungsten tip. The tip current goes into an OPA627 transimpedance amplifier with a 100 M Ω feedback resistor, in a hand-folded copper box, and is read back by a 16-bit LTC2326-16. Coarse approach is a 28BY-48 stepper on a lever. **Mechanics, controller and firmware follow Mech Panda's red-panda-stm; the scan head and preamplifier follow Dan Berard's home-built STM.** The electronics bring-up, the software, the calibration and the controls are ours.

02 The measurement chain agrees with theory

We took the tip and sample out and clipped a **known 100 M Ω resistor** where the junction would be. That makes the current something Ohm's law fixes **before the run** — so the instrument is checked against a number it had no part in choosing.



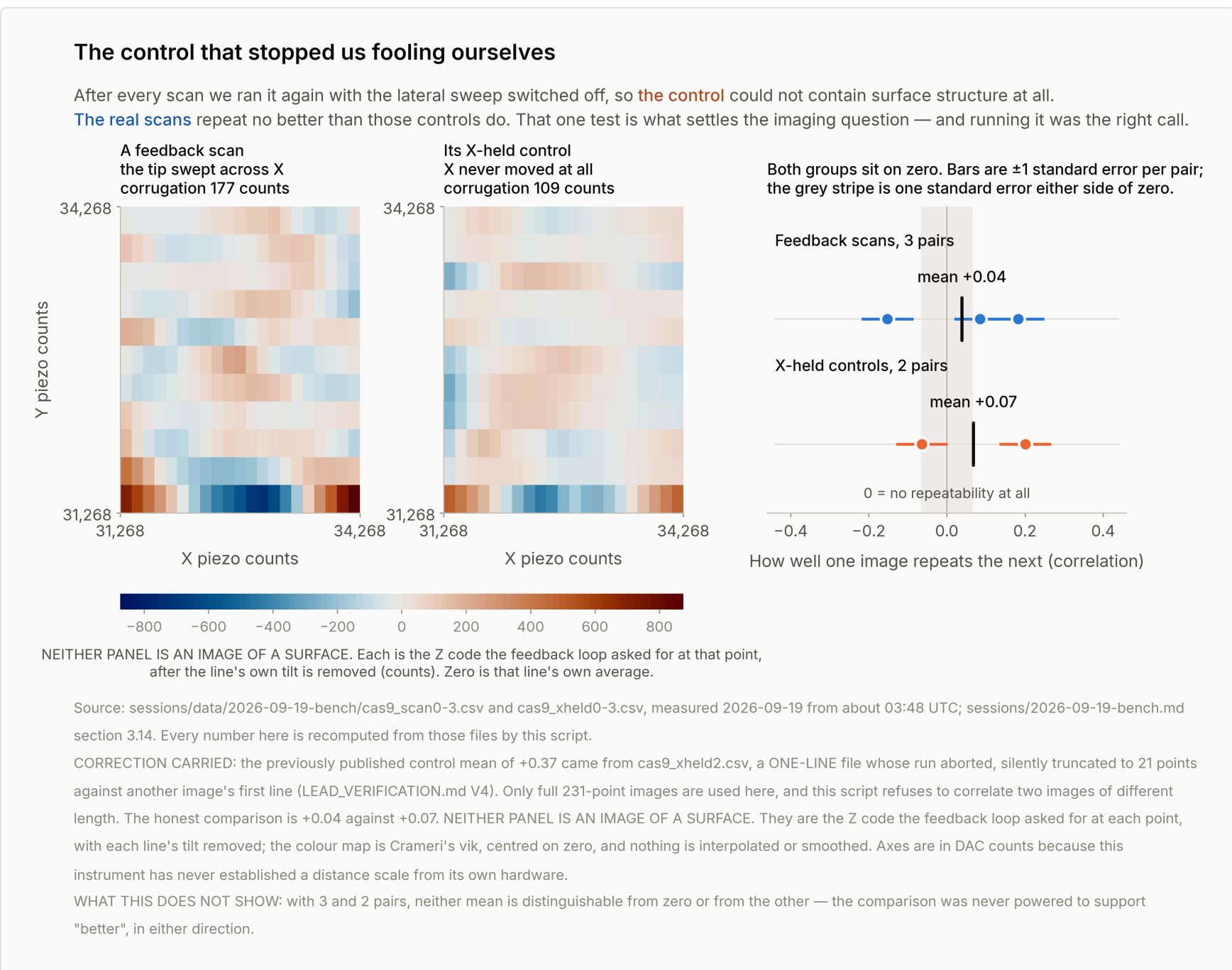
Measured $-3,204.8 \pm 36.5$ counts per volt against $-3,200$ predicted — agreement to 0.13 of one standard error, $R^2 = 0.9934$ over 53 readings. Twelve stages had to be right at once for those points to land on that line. Source: sessions/2026-09-16-bench.md §3.4. What it does not show: the slope is the ratio of two nominally 100 M Ω resistors whose tolerances were never recorded; and the 53 readings are repeats at 8 settings, not 53 independent observations.

This is the number everything else rests on

It fixes the conversion — 320.5 counts per nanoamp — and it settles the sign, which matters because a current that drove the reading the unexpected way would send an automatic approach straight into the sample.

05 We built controls designed to kill our own results

Every scan was run again with the sideways sweep switched off. An **X-held control** has identical timing and identical loop behavior, but the tip never travels across the surface — so it **cannot contain surface structure at all**, by construction.



Like for like over full images, our scans repeat at $+0.04$ and their controls at $+0.07$: indistinguishable, and neither differs from zero. That single test is what settles the imaging question, and running it was the right call. Source: sessions/data/2026-09-19-bench/ca9_".csv.

We audited ourselves, and published what we found

At this pause point we re-derived our own published numbers from the raw files. **Two statistics we had used to dismiss our own scans turned out to be computed across aborted files**, and a reproducible feature we found most convincing turned out to be the feedback loop recovering from a jump, not the sample. **Every conclusion survived. Two of the arguments did not** — and both withdrawals are in the public record.

07 The second instrument: the system that runs the science

Neither of us writes code. So the other thing we built was the framework that does — and, more to the point, the framework that catches it when it is wrong.

Two people, two computers, days apart. **The failure that kills a project like this is not a hard experiment. It is a number corrected in one document and left standing in the eleven others that quote it.**

That is not hypothetical here. On 2026-09-07 two constants changed, both were already written into more than a dozen documents, and correcting every copy anyone could think of **still left stale values in six files**.

So every number in this project now lives in exactly one place, carries a tag saying how it is known — **MEASURED, CALC, DS, MESH, ORDER, SAID, READ** — and is checked by a program, not by memory.

Where it goes next

- Measure d properly — a microscope or comparator, no power, and it settles our best junction retroactively.
- Prove the gap holds still before scanning anything — a box over the instrument, a 15-minute recording, a gate of 200 counts over sixty seconds.
- A stiffer sample than gold leaf on paper, and a sharper tip.
- Three free software fixes that remove the artefact behind our most convincing false positive: drop the first three pixels of each pass, sweep k both ways between passes, never compare two images of different length.

What we do not claim

No image. Fifty-one recordings, none of them a demonstrable image of the sample — and our own controls are how we know. **No controlled vacuum gap**, which is the regime you need in order to scan: we measured tunnelling through a barrier junction, and holding a gap open is a different thing that we have not shown. The 2026-09-19 junction was a pressed contact, measured. **No atomic resolution and no scale bar anywhere** — every axis is in converter counts, because this instrument has never established a distance scale of its own. No number is read off a photograph: captions are marked SAID or READ.

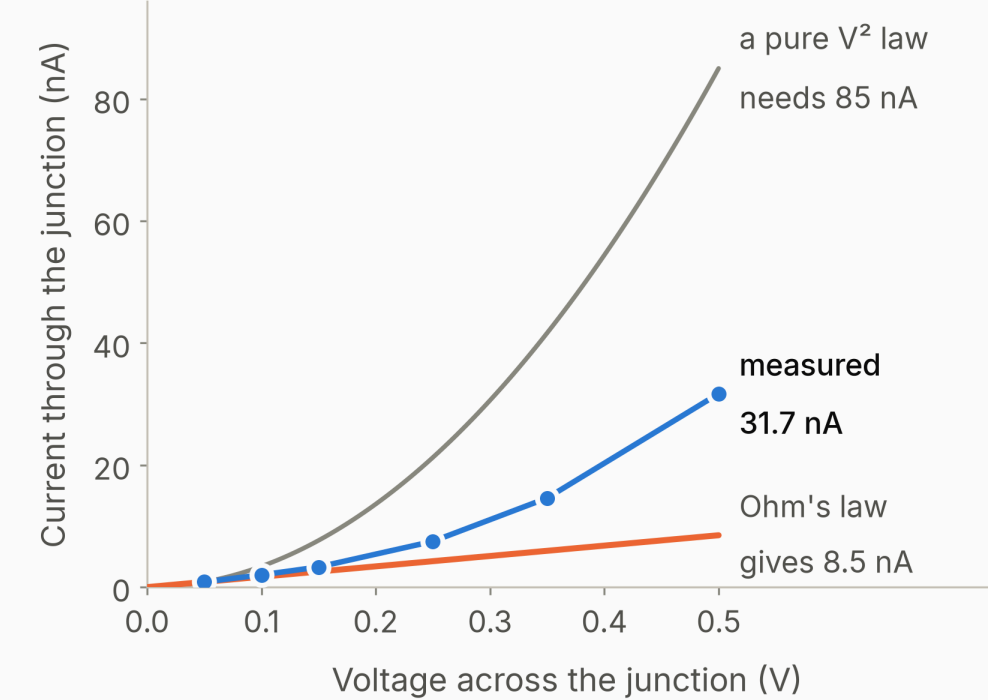
03 A real junction, with a barrier's signature

Getting a current is easy — push the tip in and there is plenty. The hard part is showing you have a **barrier**: two conductors with something in between that electrons have to cross.

We made a real tip-and-sample junction, and it behaves like a barrier

The measured current climbs far faster than the voltage does — ruling out a metallic short, which would follow the straight line of Ohm's law, and an open circuit, which would give nothing. Both polarities agreed to about 10%, interleaved so drift could not fake it.

At half a volt it passes 3.7 times the current an ordinary 59 M Ω resistor would. Fitted to the six points, current rises as $V^{1.55}$



Source: sessions/2026-09-17-bench.md section 3.15, round 7 of 8 (no CSV exists for this sweep). Measured 2026-09-17 from about 19:54:51 UTC, on the 2026-09-17 tip and sample. WHAT THIS SHOWS AND WHAT IT DOES NOT. A barrier conducting at these resistances is conducting by tunnelling, and that much is a claim this figure supports. What it does NOT show is a vacuum tunnelling gap — a contaminant film, a thin oxide or a dirty near-contact are all superlateral too, and tunnelling through a pressed film is still tunnelling but is not a gap you can command, sweep and image with. 16–59 M Ω sits at the low end of the range tunnelling occupies, which usually starts above 100 M Ω . A different tip and sample two nights later gave a different verdict (figure 5). The session log's 'roughly $V^{1.5}$ ' reading is wrong; the refit is $V^{1.55}$.

Current rose from 0.85 nA at 0.05 V to **31.7 nA at 0.5 V — 3.7 times faster than Ohm's law**, fitting $V^{1.55}$, with the junction resistance falling from 59 to 16 M Ω . Eight rounds with the polarities interleaved, so drift could not manufacture a slope; symmetric both ways to about 10%, confirmed by a bias-clip test on two separate nights. **That rules out a metallic short, which would be ohmic, and an open circuit, which would give nothing.** Source: sessions/2026-09-17-bench.md §3.15.

And the electronics are not the limit

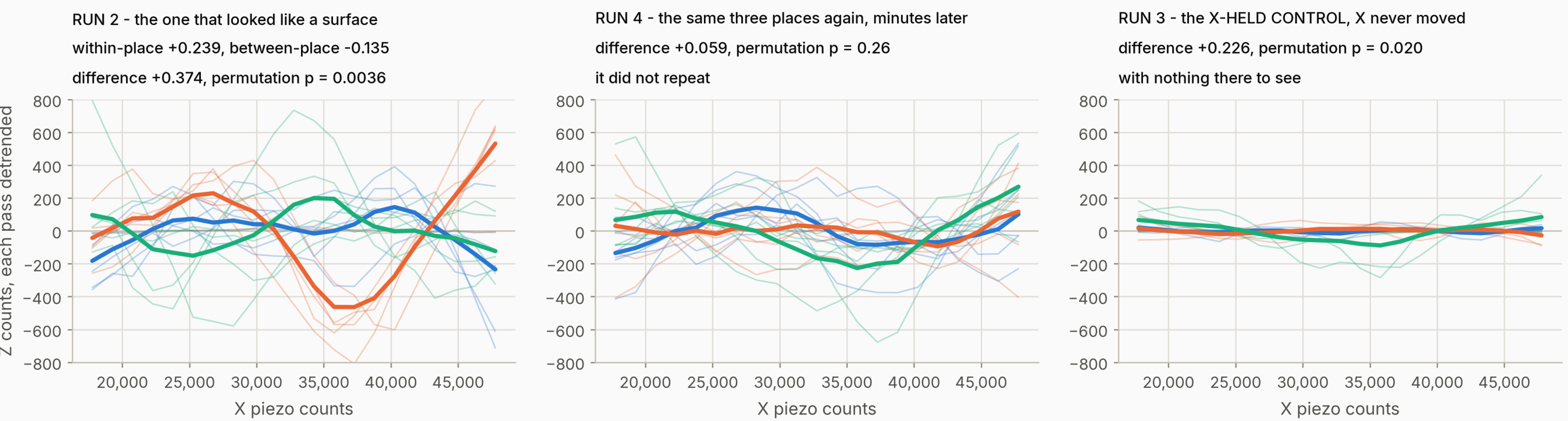
With a junction live the noise floor is **9–19 counts**; leakage through the tip lead is under 0.06 nA across ± 2 V, more than 8 G Ω ; and someone stamping on the floor two meters away moves one reading's spread from 131 to 132 pA. **The building is not getting in electrically.**

06 We detected tunnelling — and we know what stopped us holding it

Our best junction, two nights before the pressed contact, changed ten-fold about **seven times faster**: roughly 250 Z counts per decade.

The one measurement that still looks like a surface - and why it stays an open question

Six passes of one line, at three places 12,000 Y counts apart: Y $-12,000$, Y 0, Y $+12,000$. Thick lines are each place's six-pass average; thin lines are the passes themselves. Reading a surface, passes at one place would agree with each other and differ from the other two places.



Source: sessions/data/2026-09-19-bench/ycntrl_run4_3pass12000_repeat.csv, ycntrl_run4_3pass12000_repeat.csv, ycntrl_run4_3pass12000_3held.csv. Measured 2026-09-19 between 03:48 and 03:53 UTC; sessions/2026-09-19-bench.md §3.15. Each pass detrended in X, the project's own convention (Code/pc1stm3_control.py). Every file is 3 places x 6 passes x 21 points - no truncation. THE STATISTICS ARE PERMUTATION TESTS, not the 3.9 sigma reported on the night: that sigma treated overlapping pairs as independent, and was withdrawn at the 2026-09-19 pause point (Candidates/VERDICT.md 4). WHAT THIS DOES NOT SHOW: It is not an image and it is not evidence of one. Across 209 reproducibility tests on this data the honest threshold is $p < 2.4e-4$ - run 2 falls a factor of 14 short of it. Neither log carries a timestamp, so the interval between runs is UNKNOWN, bounded by the ~ 5 minutes between the sections around them (LEAD_VERIFICATION.md V9). Axes are in DAC counts; this instrument has never established a distance scale.

The one measurement that still looks like a surface. Six passes at each of three places: passes at the same place agreed $+0.239$ while different places disagreed -0.135 . The repeat, minutes later, gave $+0.056$ — and an X-held control, which cannot contain a surface, scored $p = 0.020$ on the same statistic. Against 209 reproducibility tests the honest threshold is $p < 2.4 \times 10^{-4}$. **Not an image, and not presented as one.**

We detected tunnelling. We could not hold the range long enough to scan

A tip cannot go from not touching to touching without passing through the separations where the only thing carrying electrons is tunnelling — and we had bias applied and the amplifier recording the whole way, **60,928 readings inside that range across 109 approaches**. The current is superlateral and symmetric and the junction never metallically shorted: **a barrier, conducting by tunnelling. What we have not done is hold that gap open and steady enough to scan**. That turns on d, the tip's distance from the pivot line, which sets the lever ratio: it wants to be near **26 μ m**, and we have it bounded at **under 0.5 mm**. **A microscope closes it in minutes.**

What proves it works is the retractions

- A statistical result computed across truncated files — withdrawn, and the same defect then swept across every other comparison in the project.
- Our most convincing reproducible feature — shown to be the feedback loop recovering from a flyback, predicted to the exact count. Withdrawn.
- A detector's full-scale voltage recorded backwards in a live reference document. Caught by the checker.
- Eleven wrong numbers in one commit, found by a verification pass and fixed before the second push.

An AI that writes confident prose is not hard to get. One that produces a retraction is, and that is the part we built. We did not train a model — the model is off the shelf. The discipline around it is ours.

Acknowledgements and code

Mech Panda (red-panda-stm: mechanics, controller, firmware) and Dan Berard (scan head, preamplifier) published their instruments for free. Without them there would have been nothing to bring up.

Every session log, every raw file, every analysis script and every withdrawn claim: github.com/jacobkatz/We-Are-STMING